CHAPTER 2 · DERIVATIONS

KINEMATICS

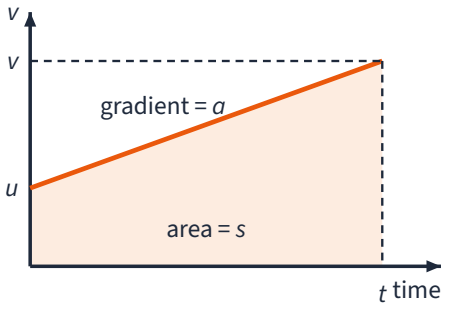
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derivation the STPM 2026 syllabus asks you to derive, deduce or show

1 EQUATIONS OF MOTION WITH CONSTANT ACCELERATION

2.1(A)

RESULT $v = u + at$, $s = ut + \frac{1}{2}at^2$, $v^2 = u^2 + 2as$, $s = \frac{1}{2}(u + v)t$

1. Constant acceleration: $a = \frac{dv}{dt}$ is constant. Integrate from $v = u$ at $t = 0$: $\int dv = \int a\,dt$, so $\mathbf{v = u + at}$ … (1)
2. $v = \frac{ds}{dt} = u + at$. Integrate from $s = 0$ at $t = 0$: $\mathbf{s = ut + \frac{1}{2}at^2}$ … (2)
3. From (1), $t = \frac{v - u}{a}$. Substitute into (2): $s = \frac{u(v - u)}{a} + \frac{\frac{1}{2}a(v - u)^2}{a^2}$.
4. Multiply by 2a: $2as = 2uv - 2u^2 + v^2 - 2uv + u^2$, so $\mathbf{v^2 = u^2 + 2as}$ … (3)
5. Area under the v–t graph (a trapezium) gives $\mathbf{s = \frac{1}{2}(u + v)t}$ … (4)



On the v–t graph, the gradient is a and the area is s.

CHAPTER 3 · DERIVATIONS

DYNAMICS

2 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 FORCE AS RATE OF CHANGE OF MOMENTUM

3.1(C)

RESULT $F = m\frac{dv}{dt} + v\frac{dm}{dt}$

1. Newton's second law: the resultant force equals the rate of change of momentum, $F = \frac{dp}{dt}$.
2. Momentum $p = mv$, so $F = \frac{d(mv)}{dt}$.
3. Product rule: $\mathbf{F = m\frac{dv}{dt} + v\frac{dm}{dt}}$.
4. Constant mass: $F = m\frac{dv}{dt} = ma$. Constant velocity (mass changing, e.g. sand falling onto a moving belt): $F = v\frac{dm}{dt}$.

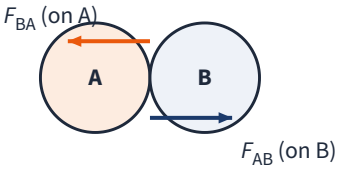
Use only the cases of constant m or constant v.

2 VERIFY CONSERVATION OF LINEAR MOMENTUM

3.2(B)

RESULT $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

1. Bodies A and B collide. During the contact time Δt, B pushes A with F_{BA} and A pushes B with F_{AB}.
2. Newton's third law: $F_{BA} = -F_{AB}$.
3. Newton's second law: $F_{BA} = \frac{\Delta p_A}{\Delta t}$ and $F_{AB} = \frac{\Delta p_B}{\Delta t}$.
4. So $\frac{\Delta p_A}{\Delta t} = -\frac{\Delta p_B}{\Delta t}$, giving $\Delta p_A + \Delta p_B = 0$.
5. $m_A(v_A - u_A) + m_B(v_B - u_B) = 0$, so $\mathbf{m_Au_A + m_Bu_B = m_Av_A + m_Bv_B}$: total momentum is conserved when no external force acts.



CHAPTER 4 · DERIVATIONS

WORK, ENERGY AND POWER

4
derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 GRAVITATIONAL POTENTIAL ENERGY NEAR THE EARTH

4.2(A)

RESULT $\Delta U = mgh$

1. Lift a mass m slowly (constant velocity, no change in KE) through a height h near the Earth's surface, where g is constant.
2. The applied force balances the weight: $F = mg$.
3. Work done $W = Fh = mgh$.
4. This work is stored as gravitational potential energy: **$\Delta U = mgh$**.

2 KINETIC ENERGY

4.2(B)

RESULT $K = \frac{1}{2}mv^2$

1. A constant resultant force F accelerates a mass m from rest through a displacement s.
2. Work done $W = Fs = (ma)s$.
3. From $v^2 = u^2 + 2as$ with $u = 0$: $as = \frac{v^2}{2}$.
4. $W = m\frac{v^2}{2}$. The work done becomes kinetic energy: **$K = \frac{1}{2}mv^2$**.

3 WORK-ENERGY THEOREM

4.2(D)

RESULT $\Delta K = F_R s$

1. A resultant force F_R changes the speed of a mass m from u to v over a displacement s.
2. $v^2 = u^2 + 2as$, so $as = \frac{v^2 - u^2}{2}$.
3. $F_R s = mas = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$.
4. **Work done by the resultant force = change in kinetic energy**, $\Delta K = F_R s$.

4 POWER

4.3(A)

RESULT $P = Fv$

1. Power is the rate of doing work: $P = \frac{dW}{dt}$.
2. For a force F along the displacement, $dW = F\,ds$.
3. $P = F\frac{ds}{dt} =$ **Fv** (if F is at angle θ to v, $P = Fv\cos\theta$).

CHAPTER 5 · DERIVATIONS

CIRCULAR MOTION

2 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 LINEAR AND ANGULAR SPEED

5.1(C)

RESULT $v = r\omega$

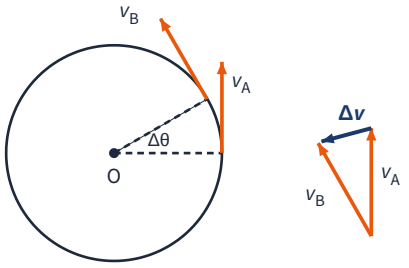
1. Arc length $s = r\theta$ (θ in radians).
2. Differentiate with r constant: $\frac{ds}{dt} = r\frac{d\theta}{dt}$.
3. **$v = r\omega$.**

2 CENTRIPETAL ACCELERATION

5.2(B)

RESULT $a = \frac{v^2}{r} = r\omega^2$

1. An object moves at constant speed v from A to B, turning through a small angle $\Delta\theta$ in time Δt.
2. The velocities v_A and v_B have the same size but different directions. The vector triangle gives the change Δv, which points towards the centre.
3. For small $\Delta\theta$ the triangle is a thin sector: $|\Delta v| = v\Delta\theta$.
4. $a = \frac{\Delta v}{\Delta t} = v\frac{\Delta\theta}{\Delta t} = v\omega$.
5. With $v = r\omega$: **$a = \frac{v^2}{r} = r\omega^2$**, directed towards the centre.



vector triangle: Δv points to O

CHAPTER 6 · DERIVATIONS

GRAVITATION

3
derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 GRAVITATIONAL POTENTIAL OF A POINT MASS

6.3(B)

RESULT $V = -\frac{GM}{r}$

1. V at a point = work done per unit mass to bring a small mass from infinity to that point.
2. At distance x the attraction on a unit mass is $\frac{GM}{x^2}$ towards M, so the external agent must pull with $\frac{GM}{x^2}$ away from M.
3. Work done per unit mass from ∞ to r: $W = \int_{\infty}^{r} \frac{GM}{x^2}\,dx = \left[-\frac{GM}{x}\right]_{\infty}^{r}$.
4. $\mathbf{V = -\frac{GM}{r}}$. It is negative because the field does work as the mass moves in; V = 0 at infinity.

2 ΔU = MGH AS A SPECIAL CASE

6.3(D)

RESULT $\Delta U = \frac{GMmh}{R^2} = mgh$

1. $U = -\frac{GMm}{r}$. Raise m from the surface (r = R) to height h (r = R + h).
2. $\Delta U = -\frac{GMm}{R+h} + \frac{GMm}{R} = GMm\,h/(R(R+h))$.
3. Near the surface $h \ll R$, so $R(R+h) \approx R^2$.
4. $\Delta U \approx GMm\,\frac{h}{R^2} = \mathbf{mgh}$, since $g = \frac{GM}{R^2}$.

3 ESCAPE VELOCITY

6.5(B)

RESULT $v_e = \sqrt{\frac{2GM}{r}}$

1. Escape velocity: the minimum speed that just takes a body to infinity, where its KE and PE are both zero.
2. Conservation of energy: $\frac{1}{2}mv_e^2 + \left(-\frac{GMm}{r}\right) = 0$.
3. $\mathbf{v_e = \sqrt{\frac{2GM}{r}}}$; at the Earth's surface $v_e = \sqrt{2gR} \approx 11.2$ km s^{-1}. It does not depend on m.

CHAPTER 9 · DERIVATIONS

DEFORMATION OF SOLIDS

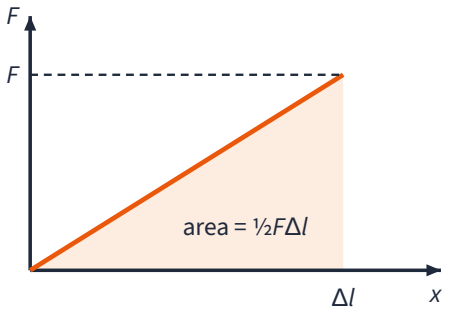
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derivation the STPM 2026 syllabus asks you to derive, deduce or show

1 STRAIN ENERGY

9.3(B)

RESULT $U = \frac{1}{2}F\Delta l$

1. Up to the proportional limit, Hooke's law: $F = kx$.
2. Work done for a small extension dx: $dW = F\,dx = kx\,dx$.
3. Total work from 0 to Δl: $W = \int kx\,dx = \frac{1}{2}k(\Delta l)^2 = \frac{1}{2}(k\Delta l)\Delta l$.
4. **$U = \frac{1}{2}F\Delta l$** = area under the force–extension graph.



CHAPTER 10 · DERIVATIONS

KINETIC THEORY OF GASES

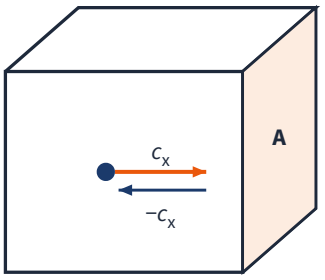
3 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 PRESSURE OF AN IDEAL GAS

10.2(C)

RESULT $p = \frac{1}{3}\rho\langle c^2\rangle$

1. N molecules, each of mass m, in a cube of side L. One molecule moves with velocity component c_x towards face A.
2. It rebounds elastically: change in momentum = $-2mc_x$, so it gives $2mc_x$ to the wall.
3. Time between hits on face A = $\frac{2L}{c_x}$.
4. Force = rate of change of momentum = $2mc_x / \frac{2L}{c_x} = \frac{mc_x^2}{L}$.
5. All N molecules: $F = (Nm/L)\langle c_x^2\rangle$; pressure $p = \frac{F}{L^2} = (Nm/V)\langle c_x^2\rangle$.
6. Random motion: $\langle c^2\rangle = \langle c_x^2\rangle + \langle c_y^2\rangle + \langle c_z^2\rangle = 3\langle c_x^2\rangle$.
7. **$p = \frac{1}{3}(Nm/V)\langle c^2\rangle = \frac{1}{3}\rho\langle c^2\rangle$**, where $\rho = Nm/V$.



face A
$\Delta p = 2mc_x$

2 MEAN TRANSLATIONAL KINETIC ENERGY

10.3(C)

RESULT $\frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$

1. From the kinetic theory: $pV = \frac{1}{3}Nm\langle c^2\rangle$.
2. Ideal gas: $pV = nRT = NkT$ ($k = \frac{R}{N_A}$).
3. $\frac{1}{3}Nm\langle c^2\rangle = NkT$, so $m\langle c^2\rangle = 3kT$.
4. **$\frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$**: the mean KE depends only on the temperature.

3 INTERNAL ENERGY AND DEGREES OF FREEDOM

10.5(B)

RESULT $U = \frac{f}{2}nRT$

1. Ideal gas: no intermolecular forces, so the internal energy is the total kinetic energy of the molecules.
2. Equipartition: each degree of freedom has mean energy $\frac{1}{2}kT$ per molecule; a molecule with f degrees of freedom has $\frac{1}{2}fkT$.
3. N molecules: $U = \frac{1}{2}fNkT = \mathbf{\frac{f}{2}nRT}$.
4. Monatomic f = 3: $U = \frac{3}{2}nRT$. Diatomic at room temperature f = 5: $U = \frac{5}{2}nRT$.

CHAPTER 11 · DERIVATIONS

THERMODYNAMICS OF GASES

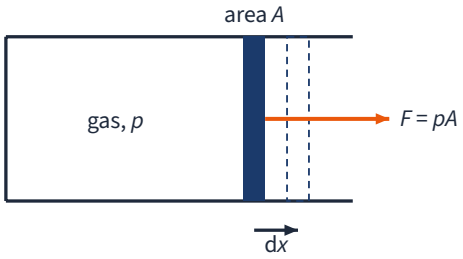
7 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 WORK DONE BY A GAS

11.2(A)

RESULT $W = \int p\,dV$

1. Gas at pressure p pushes a piston of area A with force F = pA.
2. The piston moves out a small distance dx: dW = F dx = pA dx = p dV.
3. From V_1 to V_2: **$W = \int p\,dV$** (from V_1 to V_2) = area under the p–V graph.



2 ISOBARIC WORK

11.2(B)

RESULT $W = p\Delta V = nR\Delta T$

1. Constant p: $W = \int p\,dV = p(V_2 - V_1) =$ **$p\Delta V$**.
2. $pV = nRT$ with p constant: $p\Delta V =$ **$nR\Delta T$**.

3 WORK FROM A P–V GRAPH

11.2(C)

RESULT W = area under the p–V curve

1. $W = \int p\,dV$ is the area between the curve and the V-axis from V_1 to V_2.
2. Expansion: W positive (work done by the gas). Compression: W negative.
3. For a closed cycle, the net work = area enclosed by the loop.

4 ΔU FROM AN ISOCHORIC PROCESS

11.3(D)

RESULT $\Delta U = nC_{V,m}\Delta T$

1. Isochoric: ΔV = 0, so W = 0.
2. First law Q = ΔU + W gives Q = ΔU.
3. Heat at constant volume $Q = nC_{V,m}\Delta T$, so **$\Delta U = nC_{V,m}\Delta T$**.
4. U of an ideal gas depends only on T, so this holds for any process with the same ΔT.

5 $C_{P,M} - C_{V,M} = R$ 11.3(E)

RESULT $C_{p,m} - C_{V,m} = R$

1. Heat n moles at constant pressure through ΔT: $Q = nC_{p,m}\Delta T$.
2. $\Delta U = nC_{V,m}\Delta T$ (depends only on ΔT); $W = p\Delta V = nR\Delta T$.
3. First law: $nC_{p,m}\Delta T = nC_{V,m}\Delta T + nR\Delta T$.
4. **$C_{p,m} - C_{V,m} = R$.**

6 MOLAR HEAT CAPACITIES FROM DEGREES OF FREEDOM 11.3(F)

RESULT $C_{V,m} = \frac{f}{2}R$, $C_{p,m} = \frac{f+2}{2}R$

1. $U = \frac{f}{2}nRT$, so $\Delta U = \frac{f}{2}nR\Delta T$.
2. Also $\Delta U = nC_{V,m}\Delta T$, so **$C_{V,m} = \frac{f}{2}R$**.
3. $C_{p,m} = C_{V,m} + R = \mathbf{\frac{f+2}{2}R}$.
4. $\gamma = \frac{C_{p,m}}{C_{V,m}} = \frac{f+2}{f}$: monatomic 1.67, diatomic 1.40.

7 WORK DONE IN THE FOUR PROCESSES 11.4(F)

RESULT $W = 0;\ p\Delta V;\ nRT \ln\frac{V_2}{V_1};\ \frac{p_1V_1 - p_2V_2}{\gamma - 1}$

1. **Isochoric**: $dV = 0$, so $W = 0$.
2. **Isobaric**: $W = p\Delta V = nR\Delta T$.
3. **Isothermal**: $p = \frac{nRT}{V}$, so $W = \int \frac{nRT}{V}\,dV = \mathbf{nRT\ln\frac{V_2}{V_1}} = nRT\ln\frac{p_1}{p_2}$.
4. **Adiabatic**: $pV^\gamma = K$, so $W = \int KV^{-\gamma}\,dV = \frac{K V_2^{1-\gamma} - K V_1^{1-\gamma}}{1-\gamma}$.
5. Since $KV^{1-\gamma} = pV$: $W = \mathbf{\frac{p_1V_1 - p_2V_2}{\gamma - 1}}$ $(= -\Delta U$, because $Q = 0)$.

